SEEDLINE · DESIGN PRECIS

# SeedLine design precis

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## DOCUMENT CONTROL

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| Document ID | SDL-PRC-001 | Version | 0.8 |
| Status | Draft | Date | 2026-10-24 |
| Author | Amish Chadha | License | CERN-OHL-S-2.0 |
| Project TRL | TRL 3 Proof of concept | | |
| Repository | github.com/BoujeeEnjinia1701/seedline | | |
| Source | docs/02-concept.md @ 0a57933 | | |

## REVISION HISTORY

| Version | Date | Author | Change |
|---|---|---|---|
| 0.8 | 2026-10-24 | Amish Chadha | Kit 1.9.1 upgrade: this precis describes option A of SDL-OPT-001 (concept choice Proposed, awaiting Amish); figures from SDL-CAL-001 v0.6 (fasteners weighed from the detailed model, fixes SDL-DEC-001 F1 to F8); requirements now SDL-PRS-001 |
| 0.7 | 2026-10-02 | Amish Chadha | Decisions carried into the figures: 13 mm slot with printed liners, UV-stabilized door; mass, cost and push figures from SDL-CAL-001 v0.5; plate generator usage note |
| 0.6 | 2026-10-02 | Amish Chadha | Decisions of 2026-10-02: plate generator, partner rule and first candidate, housing slot and door, no hopper window, R17 cost accepted |
| 0.5 | 2026-10-01 | Amish Chadha | Constructable design (SDL-DDR-003) and build plan (SDL-BLD-001); key numbers from SDL-CAL-001 v0.3; budget treated as a value-engineering target |
| 0.4 | 2026-09-25 | Amish Chadha | |

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|---|---|---|---|
| | | | Recommendations accepted by Amish (DDR-002) |
| 0.3 | 2026-09-25 | Amish Chadha | Record decisions (SDL-DDR-001); numbers replaced by SDL-CAL-001 results; parametric model, SDL-DWG-001 and refreshed media |
| 0.2 | 2026-09-25 | Amish Chadha | Populate to TRL 2 (architecture, metering numbers, work rate, push force, cost, safety, media) |
| 0.1 | 2026-09-24 | Amish Chadha | Initial scaffold |

# SeedLine design precis

SeedLine is a single-row push seeder. A 300 mm lugged ground wheel drives a vertical, 3D-printed seed plate through a #35 roller chain, so seeds are placed at a spacing set by distance travelled, not by walking speed. A runner opener cuts the furrow, chains cover the seed and a concave press wheel firms the row. Crop and spacing change by swapping a printed plate (1 to 36 cells) and, if needed, a 12 or 18 T wheel sprocket from an optional ratio kit. An optional telescoping marker arm scratches the line of the next row. The design is now constructable: every part can be made with a saw, drill, tap, vice and a 3D printer and bolts to the parts next to it (SDL-DDR-003), and the prototype build plan SDL-BLD-001 shows how. The TRL 3 calculations (SDL-CAL-001 v0.3) give an in-row spacing range of 22 to 589 mm with the ratio kit, a work rate of 0.142 ha/h for maize on 0.75 m rows at 2.9 km/h (7.1 h/ha against about 56 h/ha by hoe), a horizontal push of about 132 N in the design case and a mass of 14.2 kg (14.8 kg with the marker kit, SDL-CAL-001 v0.6). Value-engineering target: USD 400. Estimated cost of the constructable design: USD 261.50 with both kits (USD 138.50 under the target); the base seeder is USD 219.50, USD 19.50 over its about USD 200 target. Every requirement is met on paper or cannot be verified at TRL 3 except placement quality (R1, R2) and push effort (R10), which are at risk; mass (R11) is met with a small margin. This precis describes option A of the concept options (SDL-OPT-001); the concept choice is Proposed, awaiting Amish (SDL-DEC-001 D1).

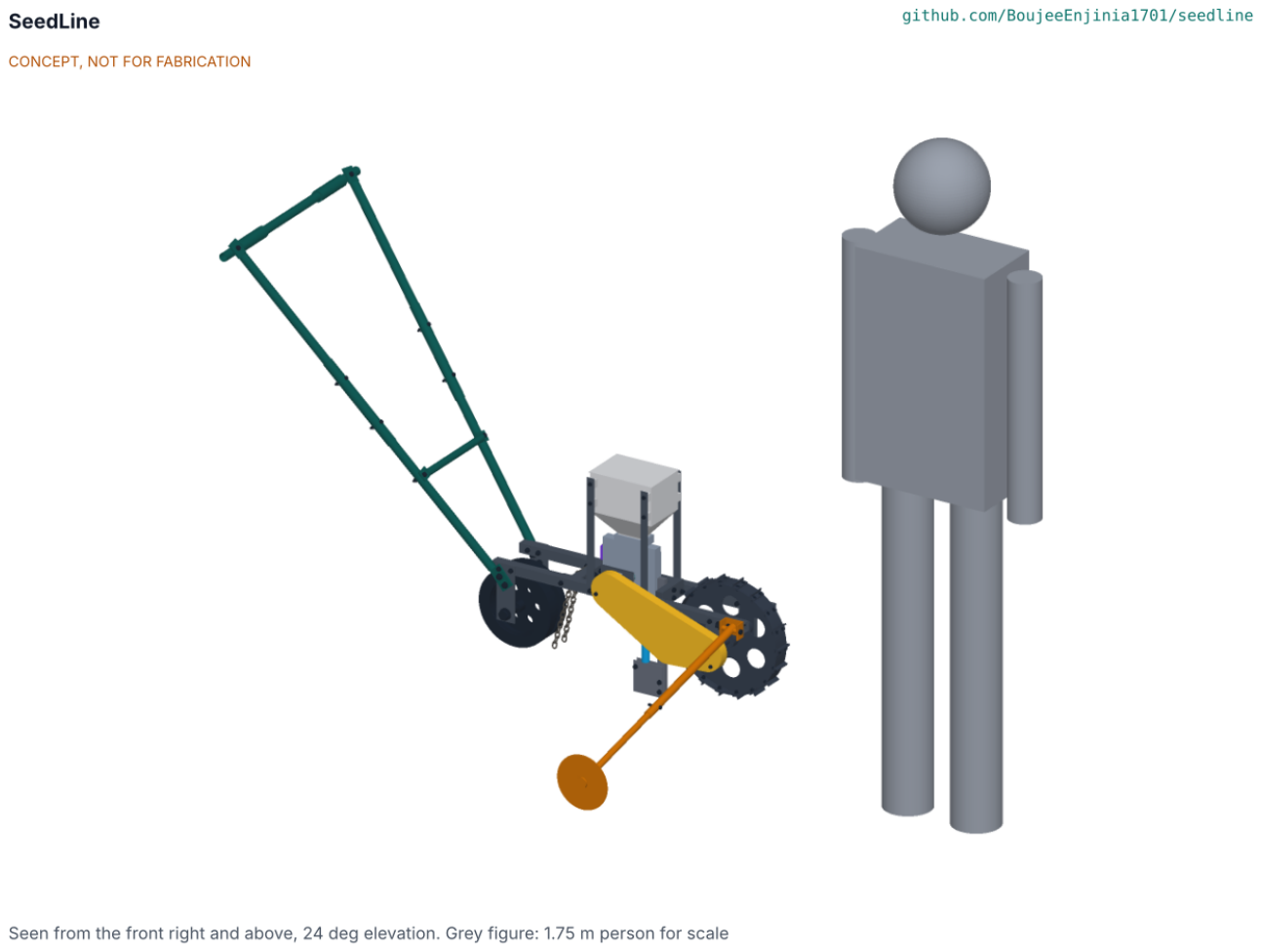



*Figure 1. SeedLine from the TRL 3 parametric model (`cad/src/model.py`), with the row marker deployed at 0.75 m, the opener at 50 mm working depth and a 1.75 m person for scale. Direction of travel is toward the lower right.*

## How it works

1. **Load.** The operator pours up to about 2 L of seed into the hopper (item 4), which feeds the metering housing (item 5) below it.
2. **Drive.** As the seeder is pushed, the lugged ground wheel (item 1) turns. A 15-tooth #35 sprocket on its axle drives a 15-tooth sprocket on the plate shaft through a 76-link roller chain (item 2), and a spring idler takes up the remaining slack. The optional ratio kit (item 16 in the BOM) adds 12 T and 18 T wheel sprockets, which change the plate speed by 0.8 or 1.2 times with a 74- or 77-link chain.
3. **Walk at the set pace.** The operator walks at about 2.9 km/h, or about 2.4 km/h with the 1.2 ratio (the walking-speed rule, SDL-DDR-002 item 5), so the cells pass the seed pool slowly enough to fill.
4. **Meter.** The printed seed plate (item 6) turns upright inside the housing. Cells on its rim pass through the seed pool at the bottom of the hopper, each picking up a seed. A brush (item 7) at the top of the plate sweeps off extra seeds, so each cell carries one.
5. **Drop.** As a cell passes the outlet on the forward side of the housing, its seed falls through the drop tube (item 8) into the furrow cut by the runner opener (item 9), just behind the opener's point.
6. **Cover and firm.** Two drag chains (item 10) pull soil over the seed and the concave press wheel (item 11) firms it, which improves seed-soil contact.
7. **Mark the next row.** The marker arm (item 14), clipped to the front cross member, drags a small disc at the set row spacing. On the return pass the operator follows that line with the ground wheel.

The frame (item 12) is two 25 x 25 x 1.5 mm square steel rails with bolted cross members, corner plates and axle drop plates; the handle (item 13), of 22 x 1.2 mm round tube, telescopes for operator height. The general arrangement is drawing SDL-DWG-001.

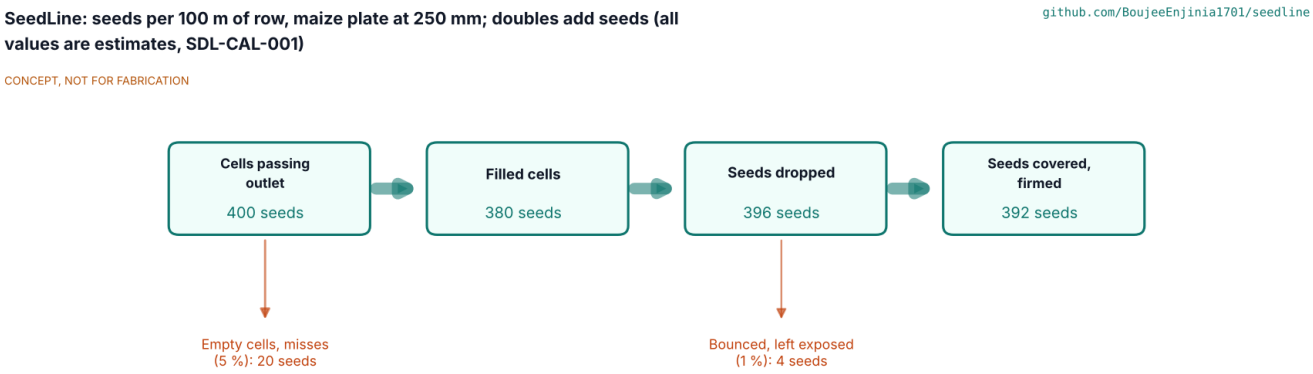



*Figure 2. Seed flow over 100 m of row for the maize plate at 250 mm spacing. All values are estimates (the miss and double rates are assumptions carried into SDL-CAL-001); doubles add seeds, so more seeds are dropped than cells are filled.*

## Main components

Numbers match the exploded view (Figure 3) and `bom/bom.csv`.

| # | Component | Proposed choice | Notes |
|---|---|---|---|
| 1 | Ground drive wheel | 300 mm diameter, 45 mm wide, 18 lugs, on a 16 mm axle with bearings | Lugs limit slip; a narrow wheel runs in the tilled row |
| 2 | Chain drive | #35 roller chain, 15 T plate and 15 T wheel sprockets, spring idler; 12 and 18 T wheel sprockets in the optional ratio kit | 76 links at 286 mm centers (74 or 77 with the kit); idler deflects up to 35 mm |
| 3 | Chain guard | Band around the chain loop with an outboard face plate, PETG or sheet steel | Covers the nip points from above, below and outside |
| 4 | Seed hopper | 2.4 L, 170 x 120 mm top on a funnel to a 60 x 30 mm throat, printed or cut from HDPE, lid | 1.73 kg of maize, 1,438 m of row per fill |
| 5 | Metering housing, shaft and bearings | Printed housing with a 13 mm slot, two printed side liners, seed outlet and a clear UV-stabilized polycarbonate side door; 12 mm shaft in two flange bearings | Side door and hand knob give tool-free plate changes; the liners fit the slot to the 6, 8 or 10 mm plate |
| 6 | Printed seed plate | 120 mm diameter, 6 to 10 mm thick, 1 to 36 rim cells sized to the seed ( `make_plate` in the model) | 40 to 54 g of PETG, 1.5 to 2.0 h to print; one plate per crop and spacing |
| 7 | Singulator brush | Nylon strip brush on a printed holder, adjustable gap | Wear part |
| 8 | Seed drop tube | 20 mm tube, housing outlet to the opener | Short drop limits bounce |
| 9 | Furrow opener and depth bracket | Runner shoe on a steel shank, clamped in a slotted bracket (10 to 60 mm depth) | Depth set against the wheels |
| 10 | Covering chains | Two short lengths of light chain on a bracket | Low cost, self-cleaning |
| 11 | Press wheel | 200 mm diameter, 70 mm wide concave rubber tread | Firms the row; carries part of the load |
| 12 | Steel frame | 25 x 25 x 1.5 mm square tube rails and cross members, 3 mm axle drop plates and bearing plate, 20 x 2 mm hopper uprights, angle corner clips (SDL-DEC-001 F6) | Bolted (decided); welding optional |
| 13 | Handle | Two 22 x 1.2 mm round tubes with a cross grip and brace, telescoping for 850 to 1,050 mm grip height | Model grip at 950 mm, 50° to the ground; 2.3 kg (lighter tube, SDL-DDR-002) |
| 14 | Row marker arm (row-spacing kit) | Telescoping arm and 140 mm marker disc, 200 to 900 mm reach, folds up for transport | Optional kit |

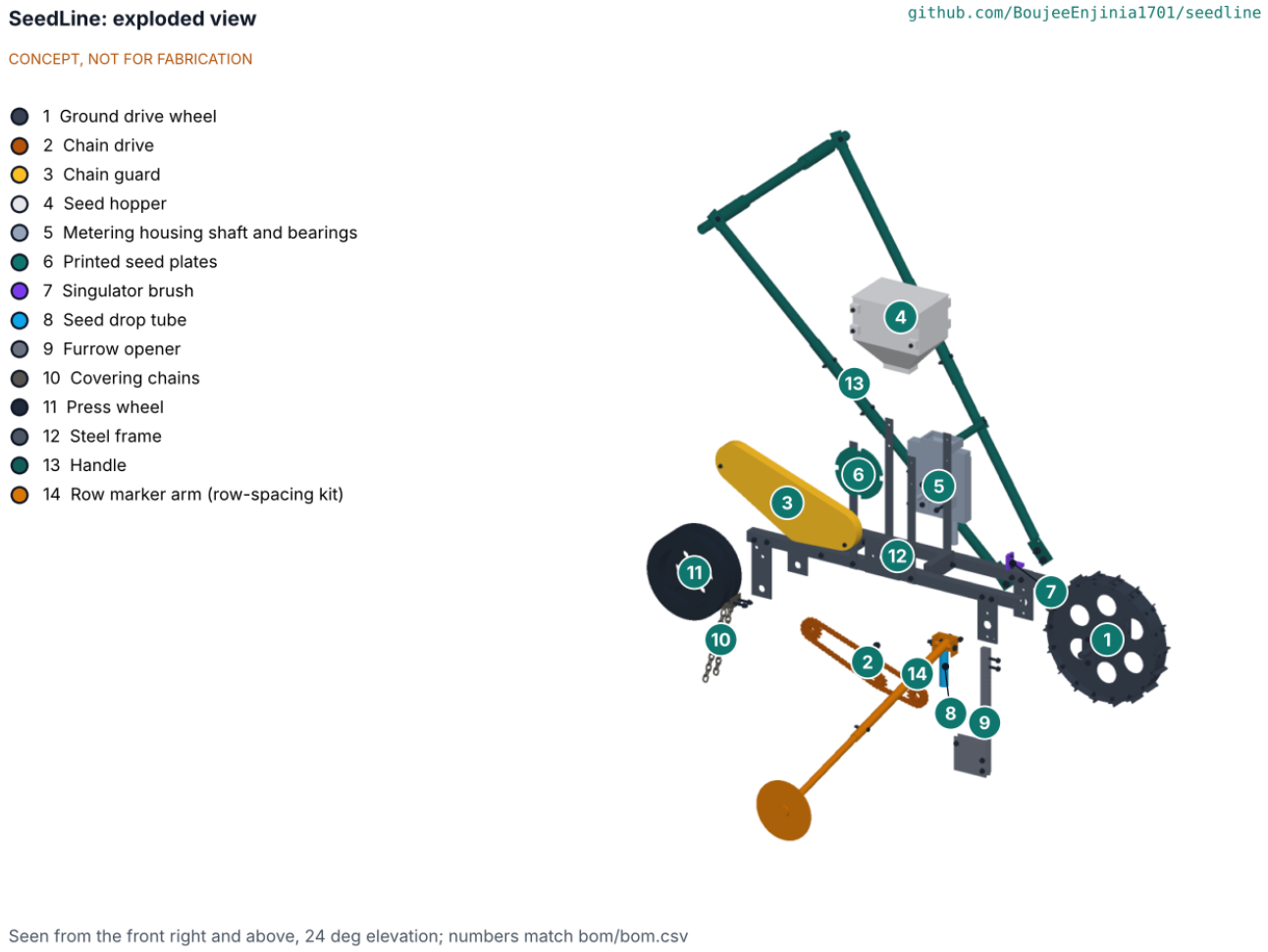



*Figure 3. Exploded view with numbered callouts matching the BOM.*

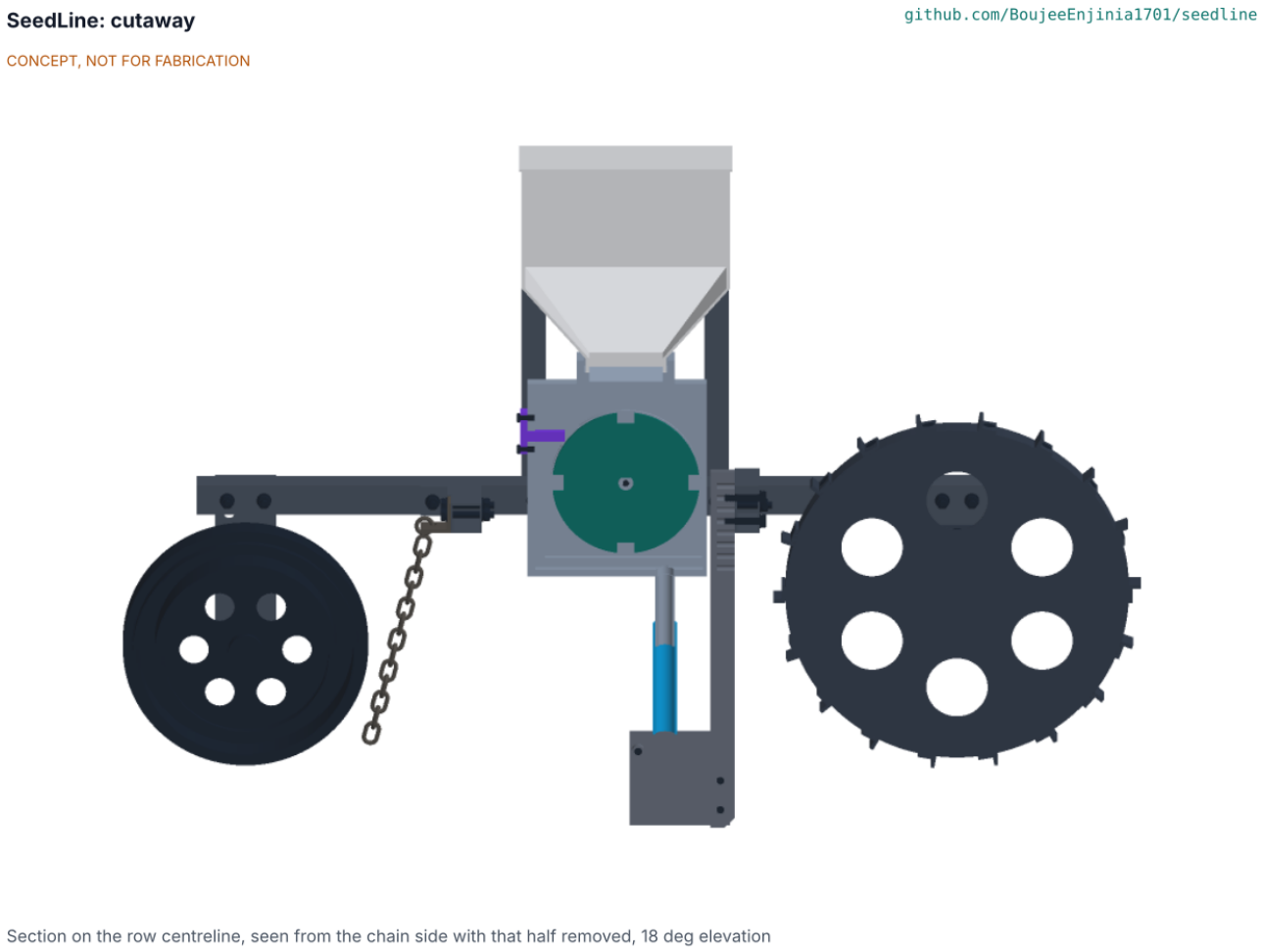



*Figure 4. Section on the row centerline, seen from the chain side with that half removed. The four-cell maize plate sits upright in the housing under the hopper funnel, with the brush at the top and the outlet feeding the drop tube into the opener. The drive wheel is on the right, the press wheel on the left.*

## Key numbers (SDL-CAL-001)

The TRL 2 estimates in version 0.2 are replaced by the calculations in SDL-CAL-001 v0.2, which reads its geometry from the parametric model. The main results:

*Table 1. Key numbers from SDL-CAL-001.*

| Quantity | Value | Requirement |
|---|---|---|
| Spacing | s = 942.5 mm / (cells x ratio), longer by 1 / (1 – slip); 22 to 589 mm nominal with 1 to 36 cells and ratios 0.8, 1.0, 1.2 (ratio kit); 26 to 471 mm with the base 15 T only | R3 met with the ratio kit |
| Maize plate | 4 cells at 1.0: 236 mm nominal, 248 mm with 5 % slip | Design case |
| Cell speed, design case | 0.299 m/s at the 2.9 km/h walking-speed rule, against an assumed 0.30 m/s fill limit | R1 at risk |
| Skid from meter torque | 1.2 to 1.7 %; rolling radius uncertain by up to 6.7 % | R4 met |
| Spacing CV, maize | 31 % of all spacings, 9 % of single spacings (assumed miss and double rates) | R2 at risk |
| Plates | 40 to 54 g PETG, 1.5 to 2.0 h each; raw seed 3.5 to 15 mm, smaller seed pelleted | R5 met, R6 met with no margin |
| Push force (horizontal part) | 81 N good, 132 N design case pushing along the handle; 179 N at best on a heavy, loose seedbed | R10 at risk |
| Mass | 14.2 kg base, 14.8 kg with the marker kit (weighed part by part, every fastener from the kit 1.9 model; mass fix SDL-DEC-001 F6) | R11 met, 0.18 kg margin |
| Work rate | 0.142 ha/h (7.1 h/ha) on 0.75 m rows at 2.9 km/h | R9 met |
| Hopper | 2.42 L, 1,451 m of maize row per fill | R12 met |
| Depth | ±10 mm needs wheel-path bumps within about ±15 mm under the drive wheel | R8 met on paper, restated for ploughed field-crop seedbeds |
| Parts cost | USD 224.10 base; USD 266.10 with the marker kit (USD 24) and the ratio kit (USD 18) | R16 USD 133.90 under its value-engineering target; R17 USD 24.10 over |

A purely horizontal push at the grip pitches the seeder forward and lifts the press wheel, so the operator pushes down along the handle; on a heavy, loose seedbed a push along the full 50° handle angle sinks the press wheel, and a flatter push of about 36° works best. The base seeder costs a little more than an Earthway 1001-B with six plates (about $187) and well under a Jang JP-1 without rollers (about $499) (UMN Extension).

With the Purdue figure of about 2.2 to 2.5 bu/acre lost per inch of spacing standard deviation (Nielsen, Purdue University), each 25 mm of spacing variation avoided is worth about 140 to 160 kg/ha of maize at those yield levels; smallholder yields are lower, so the absolute gain will be smaller.

## Key design choices

Decided by Amish on 2026-09-25 (SDL-DDR-001: go with the TRL 2 recommendations):

- **Metering type:** a vertical cell plate on a transverse shaft, driven directly by the chain. The plate is a flat print with no supports.
- **First user group:** smallholder field crops (maize, beans, sorghum, groundnut on 0.75 m rows) first; vegetable plates as a second plate set.

- **Row-spacing kit:** a telescoping marker arm. A gang bar for two or three metering units is recorded as a later variant for vegetable beds.
- **Ratio change:** plate cell count plus a 12, 15 or 18 T wheel sprocket, with a spring idler.
- **Long spacings (R3):** keep the 400 mm target and add skip-cell plates with one or two cells.
- **Small seed (R5):** pelleted seed for 1 to 2 mm seed at first.
- **Wheel layout:** front drive wheel, rear press wheel.
- **Frame joining:** bolted square tube by default; welding stays an option for workshops that weld.
- **Plate material:** PETG by default, ASA where plates sit in strong sun, PLA for trials only.

Decided by Amish on 2026-09-25 (SDL-DDR-002: go with the TRL 3 recommendations):

- **Mass (R11):** a lighter handle of 22 x 1.2 mm round tube in place of 25 x 1.5 mm, which saves 0.65 kg and brings the seeder with the marker kit to 14.9 kg.
- **Base cost (R17):** keep the target and move the 12 and 18 T wheel sprockets to an optional ratio kit; the base seeder ships with the 15 T only and costs $197.50.
- **Cell speed (R1):** a walking-speed rule of about 2.9 km/h (2.4 km/h with the 1.2 ratio); the plate size is revisited only with bench data, which is TRL 4 work and on hold.
- **Depth on rough seedbeds (R8):** accept the rigid opener for field crops on ploughed land and review with partner data; a spring-loaded opener or depth-gauge shoe is not added now.

Decided by Amish on 2026-10-02 (SDL-DEC-001):

- **Standalone plate generator.** The model's `make_plate` function already makes a plate from seed dimensions and cell count. It stays in this repo with a short usage note, and is published as a separate tool for printers only after bench tests confirm the cell sizes it produces.
- **Partner and region for co-design.** A partner already working with smallholder maize and bean farmers on 0.75 m rows on ploughed land, ideally with field trial capacity. First candidate to approach: CIMMYT's small-scale mechanization work in Eastern and Southern Africa; nothing is agreed yet.
- **Metering housing.** One housing with a 13 mm slot and printed side liners for each plate thickness, printed whole; a clear polycarbonate door in a UV-stabilized grade; no hopper window for the prototype; the steel disc drive wheel.
- **Base cost (R17).** USD 219.50 accepted for the prototype; the bronze bushing and go-kart kit savings are tried at TRL 4.

### Plate generator usage note

The function `make_plate` in `cad/src/model.py` draws a printable plate from the seed size and the cell count. Give it the seed length, width and thickness in millimetres and the number of cells; it returns a 120 mm disc with a 12 mm D-bore, a hub on the chain side and the cells cut through the rim. Cell length is 1.15 times the seed length plus 0.5 mm, cell depth is 1.05 times the seed width plus 0.3 mm, and the plate is the seed thickness times 1.2 rounded up, never under 6 mm. One or two cells make a skip-cell plate. Plates of 6, 8 and 10 mm fit the same 13 mm housing slot; the matching liner pair comes from `make_liners`. Call `python3 cad/src/model.py` to export the maize plate; for another crop, pass its seed size and cell count to `make_plate` and export the result with `export_stl`. The cell sizes are paper estimates until bench tests confirm them, and it stays in this repo until then.

## Safety

> **Safety:** SeedLine has an exposed roller chain driven by the wheel, a pointed steel opener and lugs, and is often used with chemically treated seed. Treat each as a hazard during use, cleaning, transport and plate changes.

- **Chain and sprocket nip points.** The chain moves whenever the wheel turns, including when the seeder is pushed or carried with the wheel spinning. Fingers, loose clothing and plant material can be drawn into the sprockets. The chain guard (item 3) must be fitted whenever the seeder is used, and plate changes and cleaning must be done with the wheel lifted clear and held still. Children should not push the seeder.
- **Sharp and pointed parts.** The opener point and the wheel lugs can cut shins and feet. Deburr all cut edges, round the lug tips and fit a cover over the opener for transport and storage.
- **Treated seed.** Seed dressed with fungicides or insecticides is toxic. Follow the seed label: wear gloves, do not eat, drink or smoke while filling, keep treated seed away from children and animals, and clean the hopper and housing outdoors. Never use the hopper for food or feed.
- **Row marker arm.** The deployed arm sticks out up to about 0.9 m to the side and can strike people or catch on posts at row ends. Fold it for turns near people and for transport.
- **Manual handling.** Lifting and turning at row ends and pushing on heavy soil can strain the back and shoulders (R10 and R11). Handle height must be set for the operator, and the seeder should be carried by two people or wheeled over rough ground.
- **Printed parts.** Plates, hopper and housing are plastic and can crack in cold or sun; a cracked plate can jam the drive. Inspect plates before each season and replace any that are cracked or worn.

## Open questions

TRL 4 is on hold by Amish's instruction. These questions remain for review:

- R11 is met with a small margin (0.18 kg, after the mass fix SDL-DEC-001 F6); R17 is USD 24.10 over its value-engineering target (USD 219.50 was accepted for the prototype on 2026-10-02; the plate liners and UV-stabilized door added USD 4.60). Any added part needs a matching saving.
- Whether the lighter 22 x 1.2 mm handle and its telescoping joint stand up to use (factor 1.9 on yield under the static side load; fatigue needs a test at TRL 4).
- Whether to reduce the plate diameter to lower cell speed (R1), revisited only with bench data (SDL-DDR-002 item 5).
- Whether a spring-loaded opener or a depth-gauge shoe is needed on rough, hand-tilled seedbeds (R8), to be reviewed with partner data.
- Soil data (cone index, opener draft) from the partner region to firm up the push force (R10).
- The miss and double rates of printed plates, which only a bench test can give (R1, R2).
- The first crops and spacings to support, to be set with the co-design partner (first candidate: CIMMYT's small-scale mechanization work in Eastern and Southern Africa).

Concept media: blueprint sheet, interactive 3D model. General arrangement: SDL-DWG-001. Calculations: SDL-CAL-001.